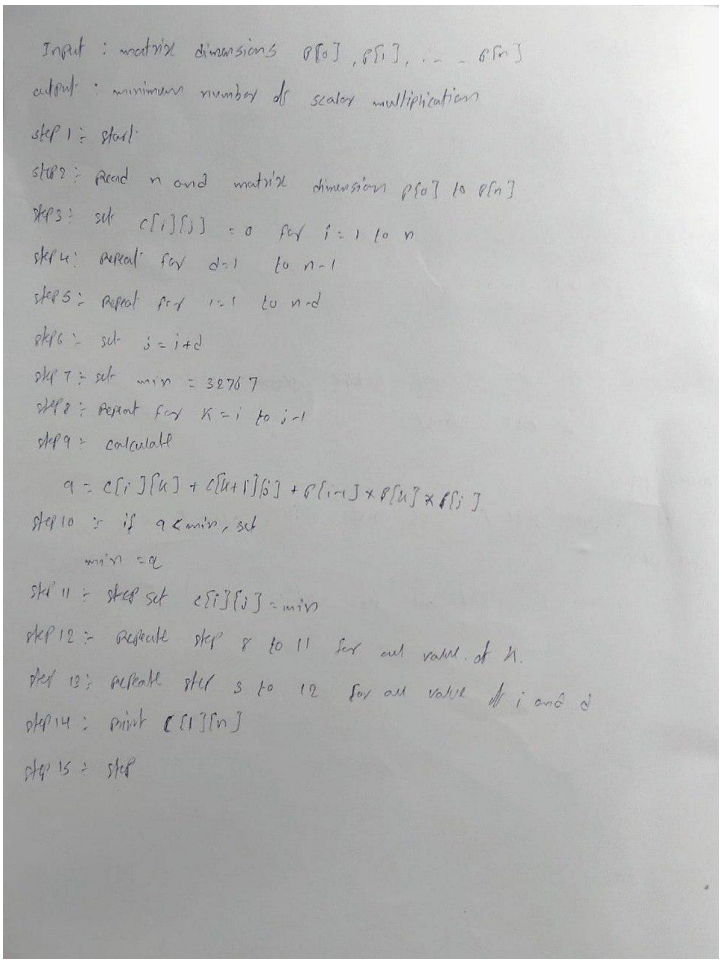
# Practical- 6

**Aim:** To Implementation of chain matrix multiplication using dynamic programming.

**chain matrix multiplication**

**Algorithm:**




**Program:**

```
#include <iostream>
using namespace std;
class MatrixChain
{
private:
   int p[20];
   int c[20][20];
   int n;
public:
   void getInput()
```

```
{
    cout << "Enter number of matrices: ";
    cin >> n;
    cout << "Enter dimensions: ";
    for (int i = 0; i <= n; i++)
    {
        cin >> p[i];
    }
}
void matrixChain()
{
    for (int i = 1; i <= n; i++)
    {
        c[i][i] = 0;
    }
    for (int d = 1; d < n - 1 + 1; d++)
    {
        for (int i = 1; i <= n - d; i++)
        {
            int j = i + d;
            int min = 32767;

            for (int k = i; k <= j - 1; k++)
            {
                int q = c[i][k] + c[k + 1][j]
                    + p[i - 1] * p[k] * p[j];

                if (q < min)
                {
                    min = q;
                }
            }
```

```
            c[i][j] = min;
        }
    }

    cout << "\nMinimum number of scalar multiplications = "
         << c[1][n] << endl;
}

void display()
{
    cout << "\nDP Table:\n";

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            if (j < i)
                cout << "0 ";
            else
                cout << c[i][j] << " ";
        }
        cout << endl;
    }
}
};
int main()
{
    MatrixChain obj;
    obj.getInput();
    obj.matrixChain();
    obj.display();
    return 0;
}
```

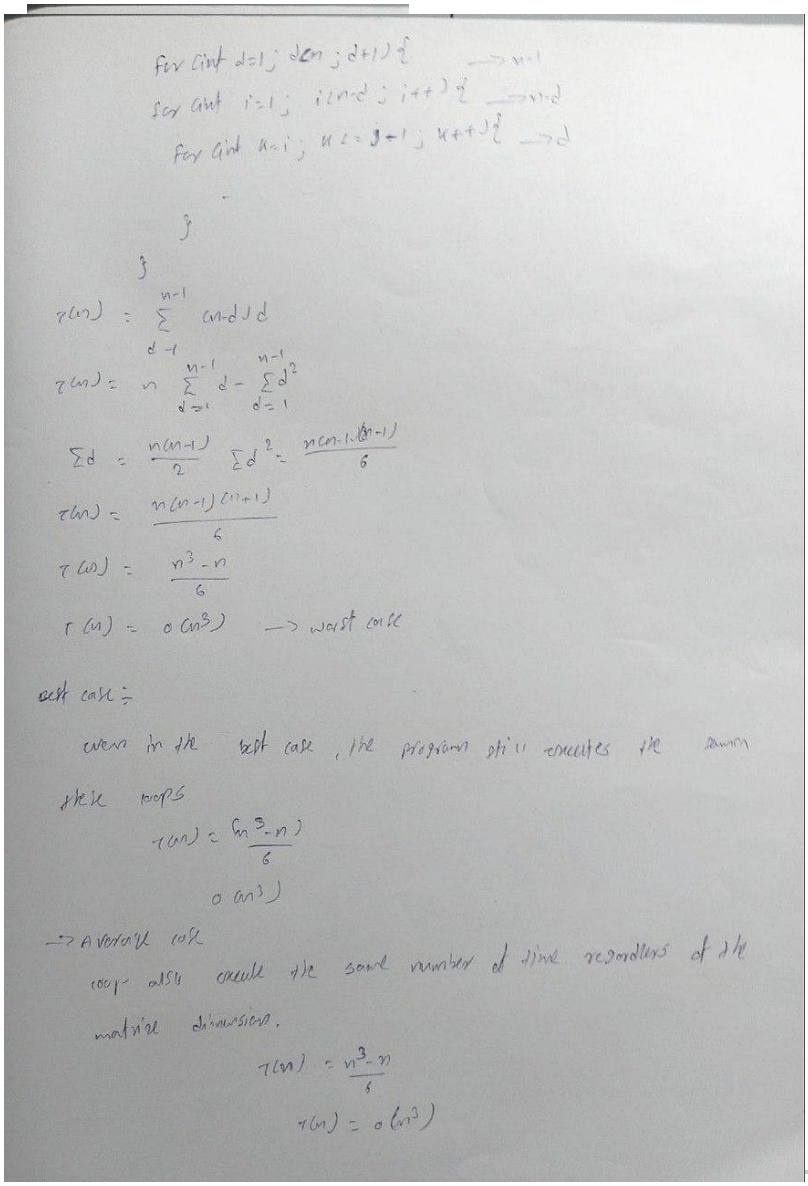
**Output:**

```
PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-6\chain matrix.exe"
Enter number of matrices: 4
Enter dimensions: 3
2
4
2
5

Minimum number of scalar multiplications = 58

DP Table:
0 24 28 58
0 0 16 36
0 0 0 40
0 0 0 0
```

**Time complexity:**




NAME:Hima sankar
Enrollment Number:92460118218

**Conclusion:** The Chain Matrix Multiplication problem was successfully implemented using Dynamic Programming. The algorithm finds the optimal order of matrix multiplication and minimizes the number of scalar multiplications. It stores the solutions of smaller subproblems in a DP table and uses them to solve the complete problem efficiently.

NAME:Hima sankar
Enrollment Number:92460118218